



Linux OS – Lab 4

Task 1: Output Redirection (> and >>)

1. Create a file named **greetings.txt** and write the text "**Hello, Linux!**" into it using **>**.
2. Append the text "**Welcome to shell scripting.**" to the same file using **>>**.
3. Display the contents of greetings.txt using **cat** command.
4. Try listing a non-existent directory and redirect the error message to **error_log.txt** file.
5. Run a command that generates an error, but discard the error output completely to **/dev/null**.
6. Run **ls** on multiple directories (Documents/, Downloads/, InvalidDir/) and save both standard output and error into a file named output.txt.
7. Use the **find** command to search for files named **shadow** inside **/etc**, saving both standard output and error to **find_results.txt**.

Task 2: Hard and Soft Links

– Hard Links

- Create a file **original.txt** with some content using **echo** command.
- Create a hard link (**hardlinkfile.txt**) using the **ln** command.
- Modify the contents of the hard link and observe changes in **original.txt**. using **cat**.
- Delete **original.txt** and verify the contents of **hardlinkfile.txt**.

– Soft Links

- Create a file **original.txt** and a soft link (**softlinkfile.txt**) using **ln -s**.
- Modify the soft link's contents and observe changes in **original.txt**.
- Delete **original.txt** and verify the behavior of the soft link.



Task 3: Managing Users in Linux

1. Create a new user using the **adduser** command with the following details:
 - ◆ **Username:** ahmed
 - ◆ **Full Name:** Ahmed Hassan
 - ◆ **Password:** 123ABC
2. Verify the user's entry in the **/etc/passwd** file.
3. Check **ahmed**'s details in the **/etc/shadow** file.
4. Update **ahmed**'s password to be **123123**.
5. Create another new user named **baduser** with a password of your choice.
6. List the available shells (i.e., **/etc/shells**) in your system.
7. Change the user's login shell to be **sh** using **usermod** command.
8. List all **basuser**'s available commands and redirect the output to **/tmp/command.list** then display their contents.
9. Delete **baduser**'s account along with the **home** directory.
10. Verify the removal of **baduser**'s via the **/etc/passwd** file.



Task 4: Managing Linux Groups

1. Create a new group named **developers**.
2. Add **ahmed** to the **developers** group.
3. Verify that **ahmed** is part of **developers** group.
4. Remove **ahmed** from the **developers** group.
5. Create another new group named **admins**.
6. Assign a password for admins group using **gpasswd** command.
7. Delete the **developers** group.
8. Use **id -gn** to view your current group.
9. Create a **main.txt** file then use **ls -l** and note the group ownership of the file.
10. Switch to a **admins** group session using **newgrp** command.
11. Use **id -gn** to view your current group session.
12. Create an **other.txt** file then use **ls -l** to view the group ownership of the file, What do you notice?
13. Execute the **exit** command (or press **CTRL+D**) to log out of the group session.



Task 5: Editing sudoers File

1. Open the **sudoers** file safely using **visudo** command.
2. Grant **ahmed** password-less sudo access to **ls** and **top** commands.
3. Add **admins** group to the **sudoers** file with access to **less**, and **ls**.
4. Add **ahmed** to the **admins** group.
5. Verify **ahmed**'s sudo permissions using **sudo -l** command.
6. Add **ahmed** to the **wheel** group, what are the effects?
7. Remove the **wheel** group from **ahmed**.